

Supplemental information

Validation of CCL20-driven CAR- $\gamma\delta$ T secreting

PD-1 blockade with enhanced

trafficking into solid tumor

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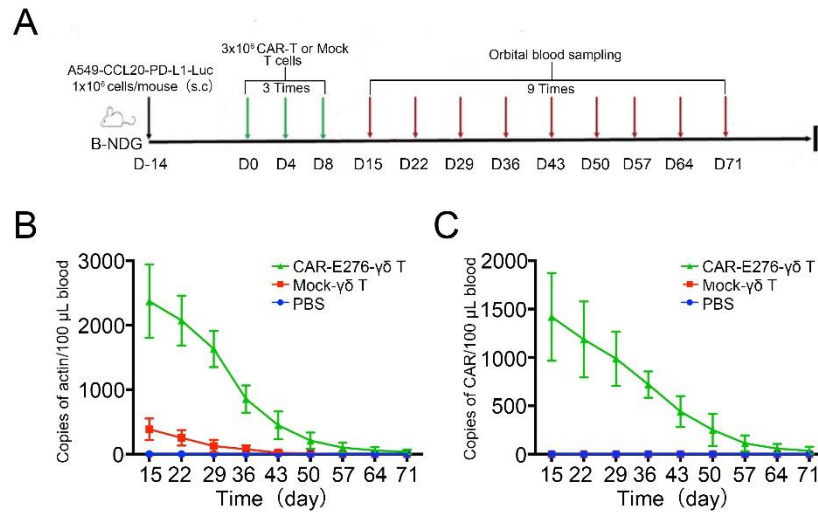


Figure S1. Persistence and Primary Pharmacokinetics profile of CAR-E276, related to Figure 4.

(A) In vivo PK study schematic for CAR-E276 in NSCLC CDX animal model.

(B and C) The persistence and pharmacokinetics of CAR-γδ T cells were evaluated in NSCLC-CDX mice (5 animals per group) in comparison to Mock-γδ T (unmodified γδ T), with PBS serving as the vehicle control. Blood samples of all animals were collected from the orbital vein once every seven days for SYBR Green-based quantitative PCR analysis to determine the copy numbers of the human actin gene and CAR-related sequences, using the corresponding primers listed in Table S3. Data are presented as mean ± standard deviation (SD) (n = 5, animals per group).

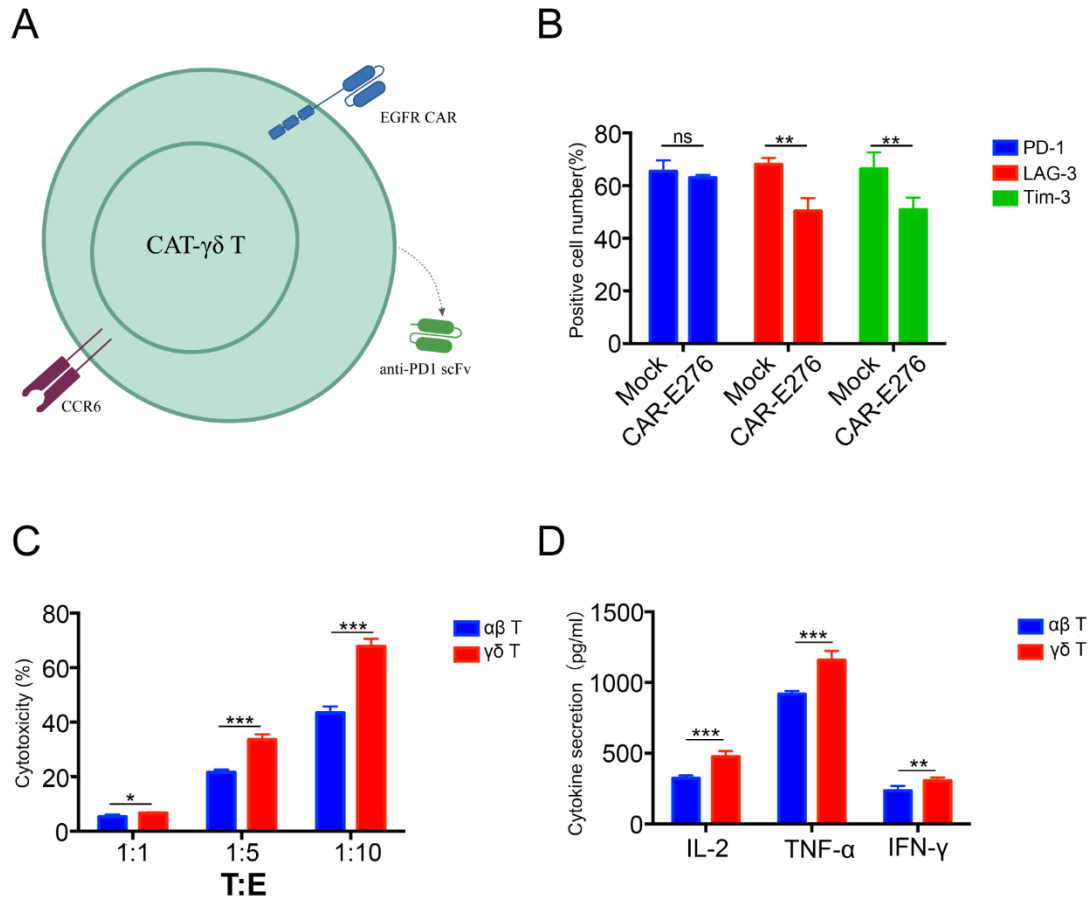


Figure S2. Structure, mechanism, and performance of CAR-E276, related to STAR method.

(A) Structural diagram of CAR-E276.

(B) The exhaustion phenotypes of CAR-E276 compared with Mock-T were detected by flow cytometry.

(C) Comparison of in vitro killing efficacy between CAR-E276 and the corresponding $\alpha\beta$ T-based CAR-T with the same modifications.

(D) Comparison of the in vitro cytokine release ability of CAR-E276 and the corresponding $\alpha\beta$ T-based CAR-T that have the same modifications.

Data are presented as mean $\pm$ standard deviation (SD) (n = 3, independent experiments). Statistical analyses were performed using the two-tailed unpaired Student's t-test. *p < 0.05, **p < 0.01, ***p < 0.001.

Comparison of groups	p Value	95% CI	Effect Size	statistical power
CAR-H VS Mock-T H	0.004	2.506~11.50	0.3454	0.890
CAR-T H VS CAR-T L	0.001	-16.40 ~-4.836	0.4232	0.968
CAR-T L VS Mock-T L	0.135	-1.824~12.63	0.1084	0.341
Mock-T H VS Mock-T L	0.007	-15.27~ -2.775	0.3121	0.828

Table S1. Statistical parameters from key groups including p value, 95% CI, effect size and statistical power based on TGI%, related to Figure 3C.

Only statistical power was calculated by using IBM SPSS Statistics 27 software, Other statistical parameters were performed by GraphPad Prism 8.0.1.

Primer	Forward	Reverse
h-CD274	5'-GGCCGAAGTCATCTGGACAA-3'	5'-GTCAGTGCTACACCAAGGCA-3'
h-CCL20	5'-GAATCAGAAGCAGCAAGCAACT-3'	5'-TTTTGGATTTGCGCACACAG-3'
h-GAPDH	5'-TCAAGGCTGAGAACGGGAAG-3'	5'-TCGCCCCACTTGATTTTGGA-3'

Table S2. PCR primer sequences for detection of overexpression of transgenes, related to STAR Methods.

h-CD274, h-CCL20 are transgenes of newly constructed cell, while h-GAPDH is a reference gene.

Primer	Forward	Reverse
CAR	5'- CTTTGGCATCGCCAACTACG -3'	5'- CTGCAGTCCGCTAATAGCCA -3'
h-Actin	5'-CACCATTGGCAATGAGCGGTTC-3'	5'-AGGTCTTTGCGGATGTCCACGT-3'

Table S3. PCR primer sequences for detection of CAR- $\gamma\delta$ T, related to STAR Methods.

Primer sequence were designed according to sequence of CAR portion expressed specifically in CAR-T; while h-actin is a house-keeping gene expressed in both CAR-T and Mock-T to comparison the persistence of CAR-T and Mock-T.